Smart Home Light Control App

Date: 23 September 2025

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1. Introduction

The purpose of this project is to design and evaluate a prototype for a Smart Home Light Control App that allows users to adjust lighting according to their mood. The project focuses on applying principles of User Experience (UX) and Usability taught throughout the course, demonstrating how emotional design can influence user satisfaction and ease of use.

Modern smart lighting systems offer extensive control over color, brightness and automation, yet many users find existing interfaces unintuitive or lacking in emotional appeal. The goal of this project was to explore how emotional design, specifically Norman's three levels of design (visceral, behavioral and reflective) can be applied to create a light control experience that evokes comfort, pleasure and personalization.

The project was conducted using a user-centered approach emphasizing users' needs and reactions throughout the design and testing phases. The outcome is a low-fidelity prototype developed in Figma that allows users to select lighting moods such as *Relax*, *Focus* or *Energize* quickly and enjoyably.

2. Methodology

The design process followed a user-centered and iterative approach, as introduced in the course. The project was divided into four parts: user research, ideation, prototyping and usability testing.

2.1. User Search

The first step was to identify who might use the app and their needs. A short interview was conducted with three participants that use smart lights at home. They expressed that the existing apps are too complex and they wanted simpler, more intuitive controls that fit specific daily moods instead of technical settings.

A persona was created to represent the target user:

Emma 27, lives in a small apartment and uses smart lights to create cozy environments. She wants to change lighting easily when studying, relaxing or having guests over.

This aligns with the user-centered design principles emphasizing early focus on users and their tasks.

2.2. Ideation and Concept Development

Using cognitive principles, the interface was designed to minimize memory load by promoting recognition over recall. Instead of requiring users to remember settings or color codes, the interface provides clear icons and mood labels such as *Relax*, *Focus* and *Energize*.

The concept also draws from external cognition: users can see the effects of their choices directly on-screen through visual previews of light color and brightness. This immediate feedback supports decision making and reinforces intuitive interaction.

Brainstorming sketches explored how to visualize moods using colors and symbols. The design goal was to make the app feel inviting and calm, rather than technical.

2.3. Prototype Creation

The prototype was developed as low-fidelity mock-up in Figma, following the prototyping principles from the course. Low-fidelity methods were chosen because they are quick to adjust and easy to test with real users.

The initial prototype (Iteration 1) consisted of three main screens:

- Home Screen containing:
    - A color preview panel at the top
    - Three main mood buttons (Relax, Focus, Energize)
    - A brightness slider at the bottom
    - A small save button centered below the brightness slider for saving the current mood
    - A menu button in the top-left corner
- Menu Screen, which appears when the user taps the menu button. It includes:
    - A button for Custom Mood
    - A button for Connect Device (not implemented)
    - A button for Settings (not implemented)
    - The same menu button in the top-left corner
- Custom Mood Screen, which allows the user to create a personal mood. It contains:
    - A color preview panel
    - A color selection panel (square color picker)
    - A color selection slider
    - A brightness slider
    - A save button in the same bottom-center location as the Home Screen
    - The menu button in the top-left corner

The design incorporated emotional design principles (Norman 2004):

- **Visceral level:** soft gradients, circular buttons, warm tones and a friendly icon set evoke positive first impressions.
- **Behavioral level:** large buttons and clear feedback make it easy to use and understand.
- **Reflective level:** The ability to name and save favorite "moods" allows users to experience identity and create meaning in their experience.

Figure 1 – Prototype Iteration 1

2.4. Usability Testing

Usability testing followed the Think-Aloud method (Olmsted-Hawala et al., 2010), where the participants were asked to perform simple tasks using the prototype while verbalizing their thoughts.

Test setup:

- 3 participants (age 23-33)
- Tasks:
    1. Change lighting to "Relax" mode
    2. Adjust brightness manually
    3. Save a custom mood named "Study Time"
- Observations: notes were taken on task completion time, errors and verbal comments.

This test provided empirical data on how users understood and emotionally responded to the prototype. Observations were used to identify usability issues and emotional responses to color and layout.

2.5. Prototype Iteration 2

After the first test, some things had to be changed in the prototype based on what people found less intuitive. The main issue was that the save button was too hidden, so in the new version it has been made bigger and with a short text next to it. Additionally, a "Custom Mood" shortcut has been added to the home screen, because one tester mentioned they expected to find it there. These changes were made to improve visibility and reduce confusion.

2.6. Usability Test Iteration 2

The updated prototype was tested again using the Think-Aloud method. This round included the same participants from the first iteration and two new participants. They performed the same tasks as in the first test. The purpose of this test was to see if the changes improved the experience.

3. Findings

3.1. Prototype Overview

The final prototype consisted of three main screens:

1. **Home Screen:** displays current light scene and quick mood buttons.
2. **Mood Selection Screen:** provides larger previews of each mood and color changes in real-time.
3. **Custom Scene Screen:** allows users to create personalized moods with color and brightness sliders.

Users responded positively to the color-coded layout and the ability to see changes immediately. All participants completed the tasks successfully.

3.2. User Test Results

Task	Success Rate	Average Time	Notes/User Feedback
Select “Relax” mode	100%	8s	“Easy to find and feels cozy”
Adjust brightness	100%	6s	“Slider works like expected”
Create custom mood	67%	20s	“Didn’t notice the save icon immediately”

Two participants described the interface as “calm” and “satisfying to use”. One noted that the icons “felt friendly”. The main difficulty was finding the save feature for custom moods.

3.3. Changes Between Iterations

Item	Iteration 1	Iteration 2	Result
Save button	Hard to find	Bigger and labeled	Both users found it immediately
Custom mood access	Only in submenu	Shortcut added	Faster access
Mood clarity	Some icons unclear	Icon and text	Users understood all mood buttons

4. Discussion

The project demonstrates how emotional design can enhance usability in smart home applications. The prototype succeeded in creating an interface that users described as simple, pleasant and personal.

At the visceral level, aesthetic choices such as soft gradients and rounded buttons immediately conveyed calmness and friendliness. This positive emotional reaction appeared to improve user confidence and satisfaction, supporting the principle that attractive things seem easier to use.

At the behavioral level, clear labels and icons helped users navigate intuitively. However, the save feature was not visually distinct enough, showing that feedback and visibility must be improved.

At the reflective level, personalization options were valued most, users enjoyed the idea of naming their own moods. This shows how self-expression and meaning can strengthen the emotional connection between user and product.

The Think-Aloud usability testing provided valuable insights, though participants occasionally needed reminders to keep talking, a known limitation mentioned in in the CHI 2010 study. Nonetheless, this method effectively revealed both usability and emotional responses.

From a cognitive perspective, the app supported recognition over recall and reduced mental effort through visual feedback. The direct mapping between mood buttons and color changes aligns with cognitive design guidelines presented during the course.

After the first round of testing, some adjustments were made to improve the prototype. The save button was made larger and given a text label because several users had difficulties finding it. A shortcut to "Custom Mood" was also added on the home screen, since one tester expected it to be there.

The second usability test showed that these changes worked: all participants were able to complete the tasks without confusion, and they found the layout easier to understand. This supports the idea that even small visual changes can have a big impact on usability. This also shows the importance of iterative design, because the first prototype often does not reveal everything.

5. Conclusion

This project applied the complete UX process to develop and evaluate a Smart Home Light Control App concept. By combining cognitive principles, emotional design, prototyping and user testing, the project demonstrated how thoughtful UX design can make technology feel more human, intuitive and emotionally engaging.

User testing confirmed that emotional design improves the perceived usability, since participants described the app as calming, simple to use and visually appealing. After the second iteration, the updated prototype performed better and users completed the tasks more easily, especially when saving custom moods.

The next step would be to create a high-fidelity interactive prototype and possibly test it together with real smart lighting hardware to see how it performs in a real environment. With more time, a third iteration could also explore more advanced features, such as animations or more mood categories.

Reflecting on the process, this project provided hands-on experience with the UX process: starting from user research, moving through low-fidelity prototyping, testing and reflecting on improvements. The second iteration showed clearly why UX work is never finished after the first design and how each round of testing helps move the design closer to what users need.

6. Appendix

This could contain essential parts of audio/video capture of participatory design workshops, usability tests, prototype presentations, surveys, personas, etc….